

Vasu Mekala

United States | vasusys111@gmail.com | (213) 876-9232 | [LinkedIn](#) | [Website](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of experience building machine learning solutions across banking, telecom, healthcare insurance, and pharmacy analytics environments. Strong background in developing models for credit risk, fraud detection, churn, patient risk, forecasting, segmentation, and customer intelligence, with a focus on practical business use and measurable outcomes. Experienced across the machine learning lifecycle, from data preparation and feature engineering to model training, evaluation, explainability, monitoring, and production-readiness support. Works closely with risk, compliance, healthcare, marketing, engineering, and analytics teams to turn complex data into reliable AI/ML solutions that support decisions, reduce manual effort, and improve operational visibility.

EDUCATION

University of New Haven
Master of Science, *Data Science*

Aug 2023 - Dec 2024
GPA: 3.8

SKILLS

Programming Languages: Python, SQL, R

Frameworks & Libraries: PySpark, Pandas, NumPy, SciPy, Scikit-learn, XGBoost, PyTorch, TensorFlow, Hugging Face Transformers, BERT, LangChain, LlamaIndex, FastAPI

Tools & Software: MLflow, AWS SageMaker Pipelines, Model Registry, Model Deployment, Model Serving, Model Monitoring, Drift Detection, Automated Retraining, CI/CD for ML, Docker, Apache Airflow, Databricks, SHAP, LIME, SR 11-7 Documentation, HIPAA, ROC-AUC, Precision-Recall Analysis, F1 Score, Calibration Checks, Population Stability Index, Data Quality Validation, A/B Testing, Bayesian Inference, Hypothesis Testing, Econometric Regression, Time-Series Forecasting, ARIMA, Prophet

Cloud Platforms: AWS, AWS SageMaker

Soft Skills: Collaboration, Validation, Documentation, Stakeholder Engagement

WORK EXPERIENCE

Capital One | AI/ML Engineer - Consumer Banking, Credit Risk & Fraud AI Jun 2025 - Present

- Developed credit-loss forecasting models with Python, Scikit-learn, XGBoost, econometric regression, macroeconomic indicators, and portfolio-risk features, raising forecast stability by 16% for stress-testing and risk-planning cycles.
- Shaped model-ready datasets from credit utilization, payment behavior, delinquency trends, exposure limits, transaction activity, and customer-risk attributes using Python, SQL, Pandas, and PySpark to support cleaner risk and fraud model training.
- Built PyTorch fraud-detection components with Hugging Face Transformers, behavioral embeddings, merchant-category sequences, amount velocity, and transaction timelines to catch suspicious banking patterns beyond rule-based controls.
- Tuned fraud-model thresholds through ROC-AUC, precision, recall, F1 score, class-imbalance handling, and cross-validation, decreasing unnecessary investigation alerts by 12% while protecting high-risk transaction coverage.
- Applied SHAP and LIME explainability across credit-risk, fraud, and credit-limit propensity models to turn complex predictions into reason codes, feature-impact summaries, and Responsible AI evidence for validation and compliance teams.
- Configured AWS SageMaker Pipelines, MLflow, Model Registry, drift detection, population stability index, and prediction-monitoring checks across 6 production ML workflows to help risk teams catch model degradation earlier.
- Automated retraining and model-review triggers using model monitoring, calibration checks, CI/CD for ML, Docker, Apache Airflow, and FastAPI handoffs, cutting manual monitoring review effort by 22%.
- Partnered with model validators, fraud strategists, data engineers, and compliance teams to document assumptions, limitations, validation results, monitoring plans, and approval evidence for auditable production ML decisions.

Verizon | Machine Learning Analyst - Consumer Analytics & AI Nov 2024 - May 2025

- Built postpaid churn models with Python, Scikit-learn, XGBoost, billing behavior, usage signals, and service-interaction data, lifting retention model accuracy by 15% for consumer analytics decisions.
- Unified 500K+ call-detail records, network events, plan attributes, payment activity, and campaign-response data using SQL, PySpark, Pandas, and Databricks to create reliable ML training datasets.
- Modeled uplift and propensity patterns with classification, feature engineering, and treatment-response variables, helping retention teams focus offers on customers most likely to respond.
- Produced churn scores, customer segments, and treatment recommendations through Python, XGBoost, model validation, and feature-importance analysis to guide loyalty and service-recovery campaigns.
- Interpreted NPS and satisfaction drivers with Bayesian inference, hypothesis testing, regression analysis, Pandas, and NumPy, increasing visibility into customer pain points by 18%.
- Validated retention experiment outcomes using A/B testing, Bayesian inference, confidence intervals, and segment-level analysis so marketing teams could choose campaign actions with stronger evidence.
- Streamlined Databricks and PySpark feature pipelines for network-event logs, usage shifts, dropped-call patterns, support frequency, and payment behavior, cutting recurring preparation effort by 35%.
- Reviewed model performance through ROC-AUC, precision-recall analysis, calibration checks, feature importance, and MLflow tracking to support production-ready recommendations with marketing and network teams.

HCL Technologies | Data Scientist - Healthcare Insurance Analytics Feb 2022 - Aug 2023

- Developed patient readmission-risk models with Python, Scikit-learn, PyTorch, diagnosis history, medication activity, and utilization signals, giving care teams earlier visibility into high-risk member groups.
- Curated ML training datasets from 2M+ claims, encounters, admissions, eligibility records, discharge details, and pharmacy data using SQL, PySpark, Pandas, and NumPy for dependable healthcare model development.
- Shaped clinical and claims-based features around emergency visits, chronic-condition burden, length of stay, prior admissions, care gaps, and medication patterns, strengthening readmission-risk scoring for U.S. payer analytics.
- Applied SHAP explainability to readmission and utilization models with Python, translating complex prediction outputs into risk drivers that clinical, operations, and governance teams could review clearly.
- Built ARIMA and Prophet forecasting models with Python, seasonal trends, arrival patterns, and historical utilization data, lowering average emergency-room wait time by 18% through better capacity planning.
- Validated classification and forecasting outputs through regression checks, feature-importance review, model validation, and segment-level analysis, keeping predictions consistent across 6 patient-risk segments.
- Automated recurring claims and billing reconciliation checks with Python, SQL, Pandas, and data-quality validation, saving 20+ weekly manual hours across healthcare reporting workflows.
- Collaborated with healthcare stakeholders, business analysts, data engineers, and QA teams on HIPAA-aligned validation reviews, supporting cleaner analytics releases for care coordination and payer reporting.

Cognizant | Programmer Analyst Trainee - Healthcare Analytics & ML Support Oct 2020 - Jan 2022

- Built customer lifetime value segmentation models with R, regression, K-Means clustering, prescription activity, and engagement signals, increasing targeted health-program enrollment by 14%.
- Cleaned enrollment, adherence, prescription, and customer activity datasets using SQL, Python, Pandas, and NumPy so ML analysts had dependable inputs for pharmacy campaign analysis.
- Modeled member groups by expected value, medication adherence, program participation, and purchase behavior to help healthcare teams personalize outreach for U.S. pharmacy programs.
- Converted segmentation results into campaign-ready customer lists with Python, SQL, and feature engineering, giving operations teams clearer targets for cross-sell and enrollment planning.
- Automated recurring compliance and operations checks with Python, SQL, Pandas, and validation rules, shortening pharmacy analytics report preparation from 3 weeks to 2 days.
- Orchestrated scheduled pharmacy data feeds with Apache Airflow and Docker workflows, sustaining 99.9% SLA compliance for recurring healthcare analytics jobs.
- Verified customer, prescription, and enrollment records through SQL and Python checks, catching duplicate transactions, missing identifiers, and incomplete periods before downstream analysis.
- Partnered with analysts, developers, QA teams, and pharmacy-program stakeholders to resolve data issues and deliver 20+ recurring analytics outputs for business users.

PROJECTS

Enterprise Generative AI Knowledge Assistant

- Designed a LangChain and LlamaIndex-based RAG assistant to search policy, technical, and operational documents through Vector Search, giving users grounded answers with retrieved source context.
- Built Python pipelines for document chunking, metadata tagging, semantic retrieval, prompt engineering, and FastAPI response delivery, making unstructured enterprise content easier to query and validate.
- Tracked prompt versions, retrieval quality, response relevance, and LLM Evaluation results in MLflow, supporting safer GenAI application testing before production deployment.

NLP Claim Review and Member Communication Intelligence

- Fine-tuned BERT models with PyTorch and Hugging Face Transformers to classify claim notes and member-service text into clinical, billing, eligibility, and service-related categories.
- Extracted entities from unstructured healthcare communication using Named Entity Recognition, embeddings, and Python text preprocessing, giving reviewers clearer signals from notes that were difficult to analyze manually.
- Evaluated NLP outputs through precision-recall analysis, classification reports, and segment-level error review, building stronger trust in text classification results for downstream healthcare analytics use.

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Specialization - Coursera
- AI and LLM Engineering - Coursera
- IBM AI Engineering Professional Certificate - Coursera
- Generative AI with Large Language Models - Coursera